



Mechanisms of Antimicrobial Resistance and Virulence

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Learning Objectives

By the end of this module, participants will be able to:

- ✓ Define and Explain AMR
- ✓ Describe the Major Mechanisms of AMR
- ✓ Integrate Genomic Insights
- ✓ Define virulence factors and their importance
- ✓ Describe categories and molecular basis of virulence
- ✓ Explore VFDB and related tools/databases
- ✓ Perform hands-on virulence gene identification



Antimicrobial Genes & Mechanisms

Overview of AMR

- **Definition:** Antimicrobial Resistance (AMR) is the ability of microorganisms to resist the effects of medications that once effectively treated them.
- **Global Impact:** AMR is a significant public health threat, exacerbated by the overuse of antimicrobials in humans and agriculture (Waddington *et al.*, 2022).

Key Statistics:

- Responsible for millions of deaths annually.
- Estimated economic burden exceeding \$1 trillion by 2050 if not addressed (Hendriksen *et al.*, 2019)

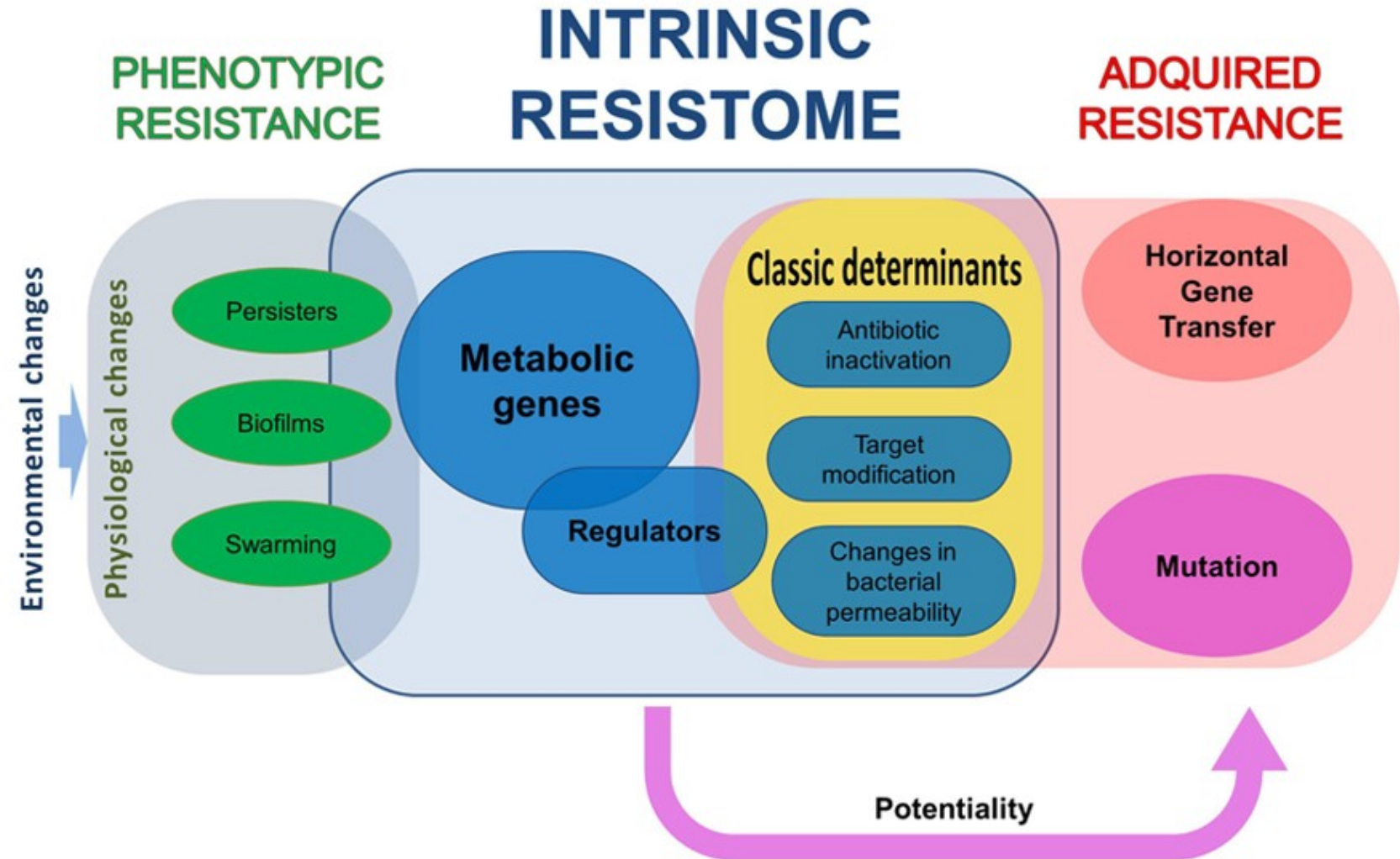
Intrinsic vs Acquired Resistance

Intrinsic resistance:
natural, species-specific.

- Impermeable cell walls, efflux pumps, lack of drug target

Acquired resistance:
mutations or HGT.

- Driven by plasmids (e.g., tet, qnr, aac(6')-Ib-cr), integrons, and transposons



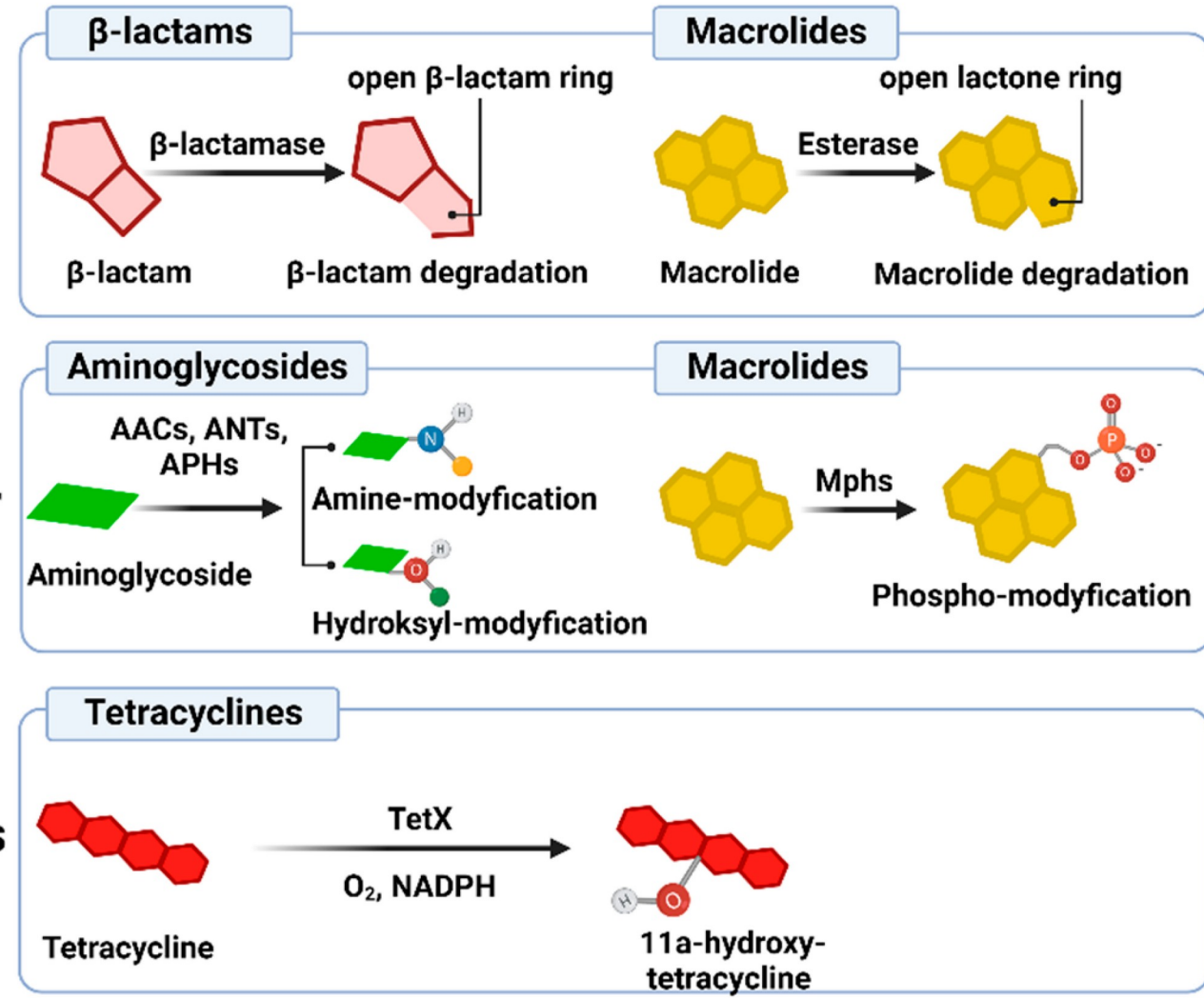
Mechanism 1: Enzymatic Inactivation

- Bacteria produce enzymes that degrade or modify antibiotics.
- Examples:
 - β -lactamases (ESBL, carbapenemases: NDM, KPC, VIM, OXA, IMP).
 - Aminoglycoside-modifying enzymes.

1 Hydrolysis

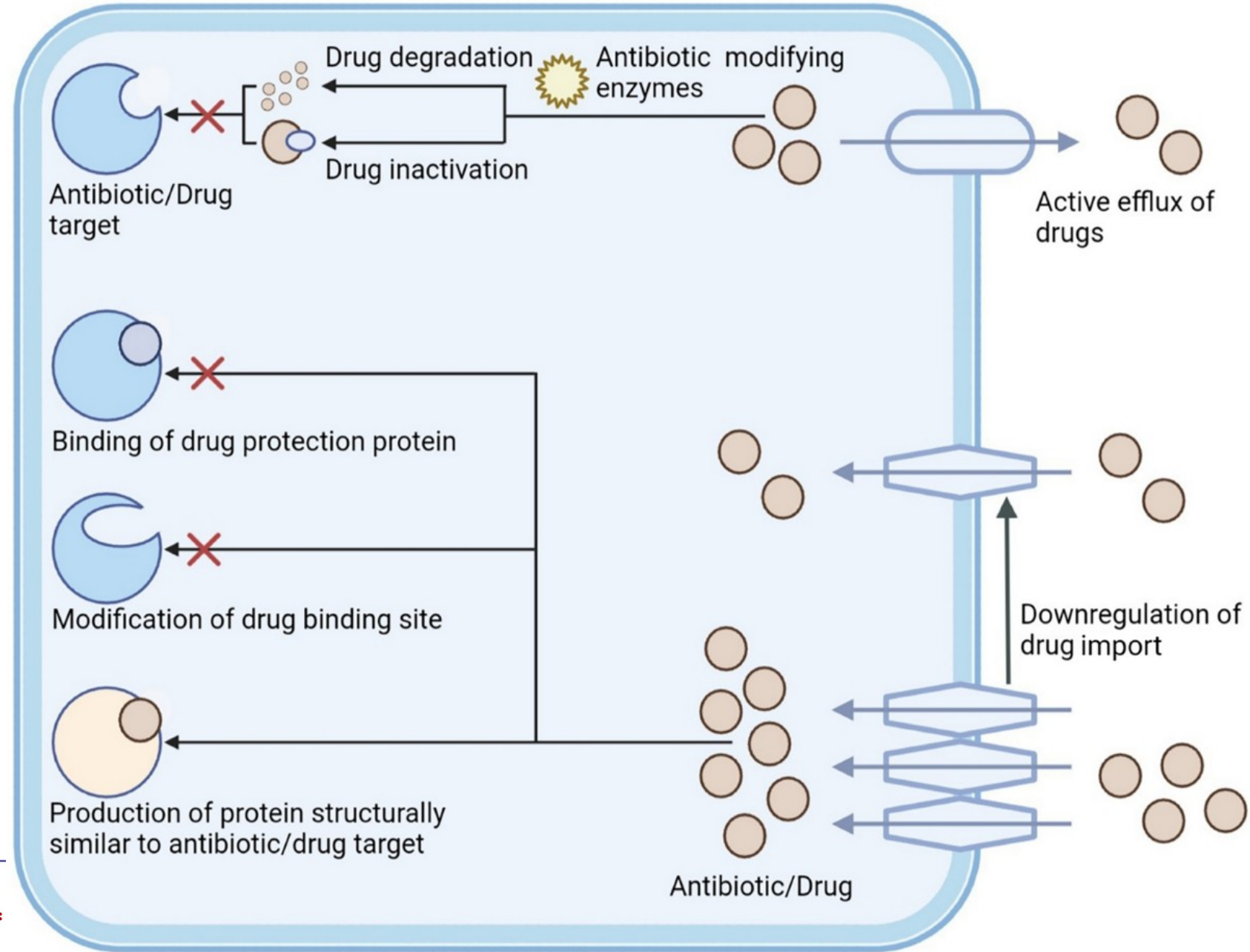
2 Group transfer

3 Redox process



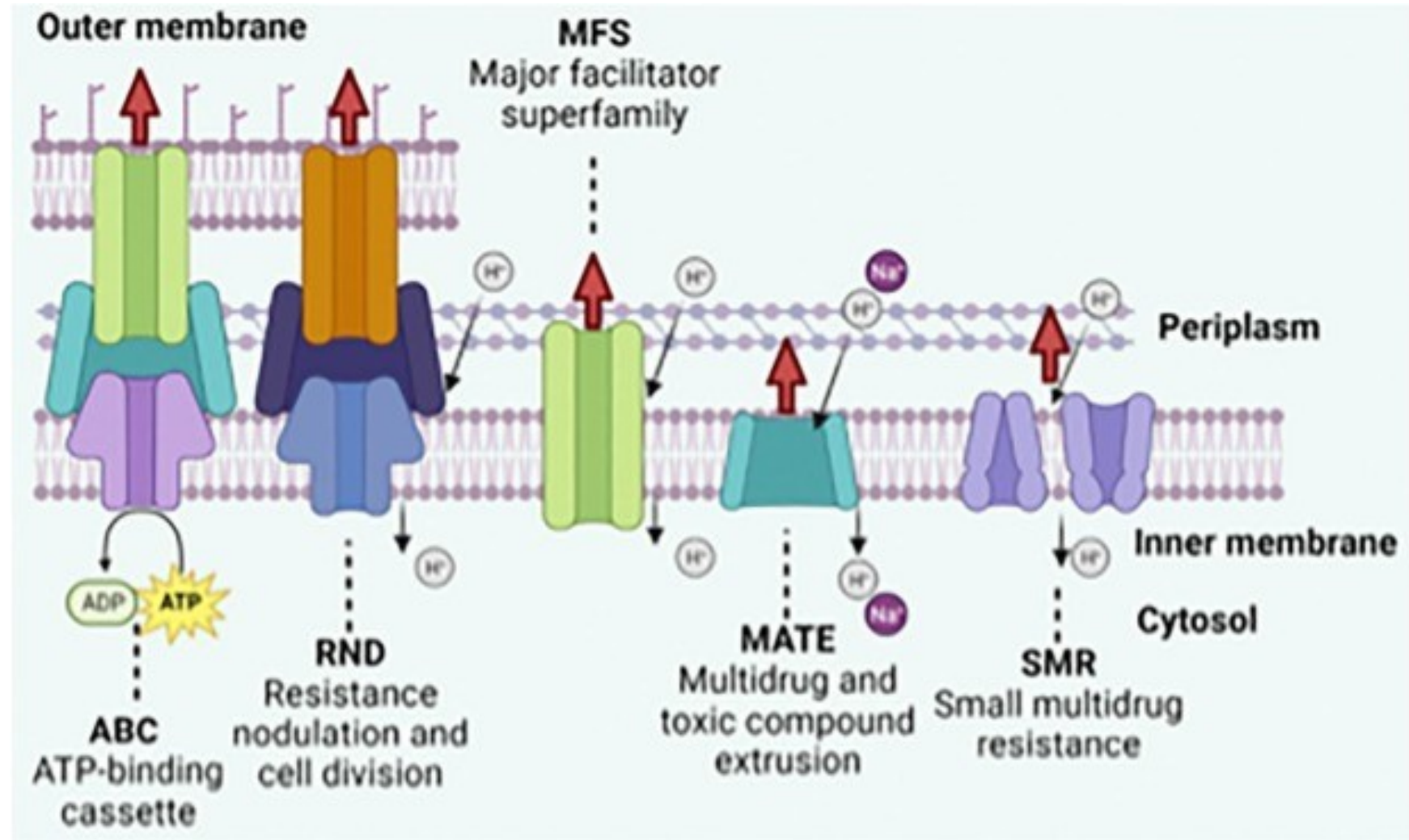
Mechanism 2: Target Modification

- Structural changes prevent antibiotic binding.
- Examples:
 - Altered PBPs (mecA/mecC → MRSA).
 - Ribosomal RNA methylation → macrolide/tetracycline resistance.
 - DNA gyrase/topoisomerase mutations → quinolone resistance.



Mechanism 3: Efflux Pumps

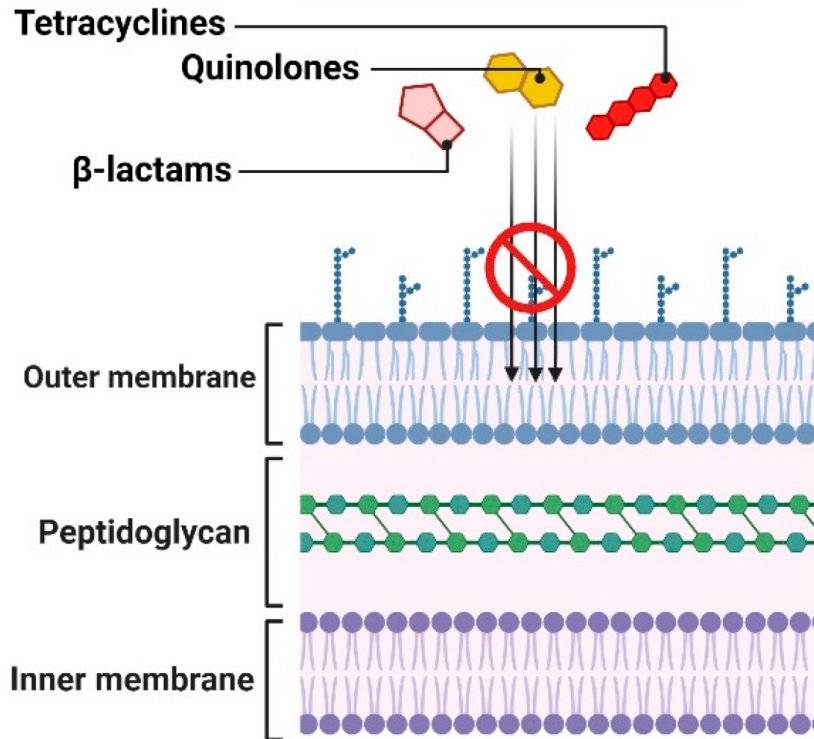
- Energy-dependent export of drugs.
- Families: RND (AcrAB-TolC), MFS, SMR, MATE.
- Plasmid-borne efflux genes: oqxAB, qepA
- Contributes to multidrug resistance.



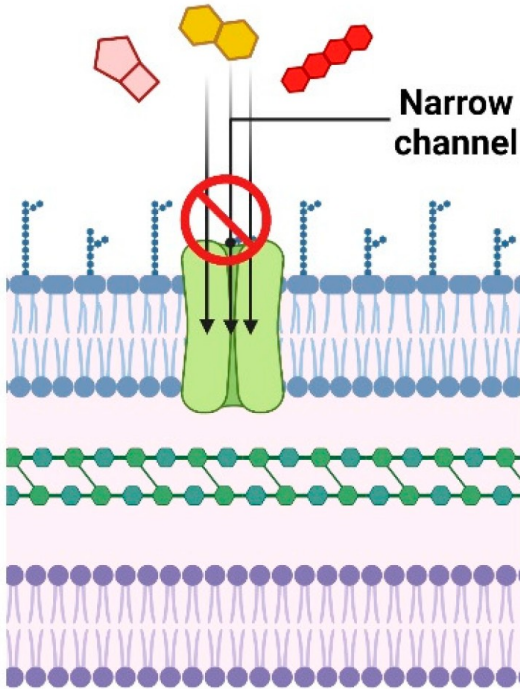
Mechanism 4: Reduced Permeability

Changes in the permeability of bacterial cell

Porin loss



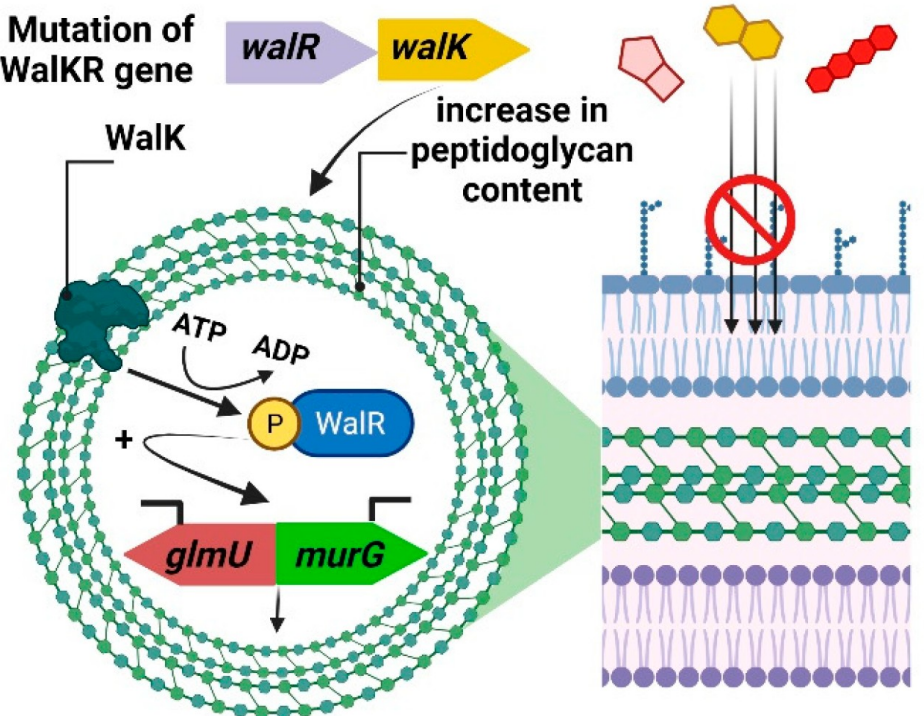
Porin mutation



Mutations of WalK/R two-component system

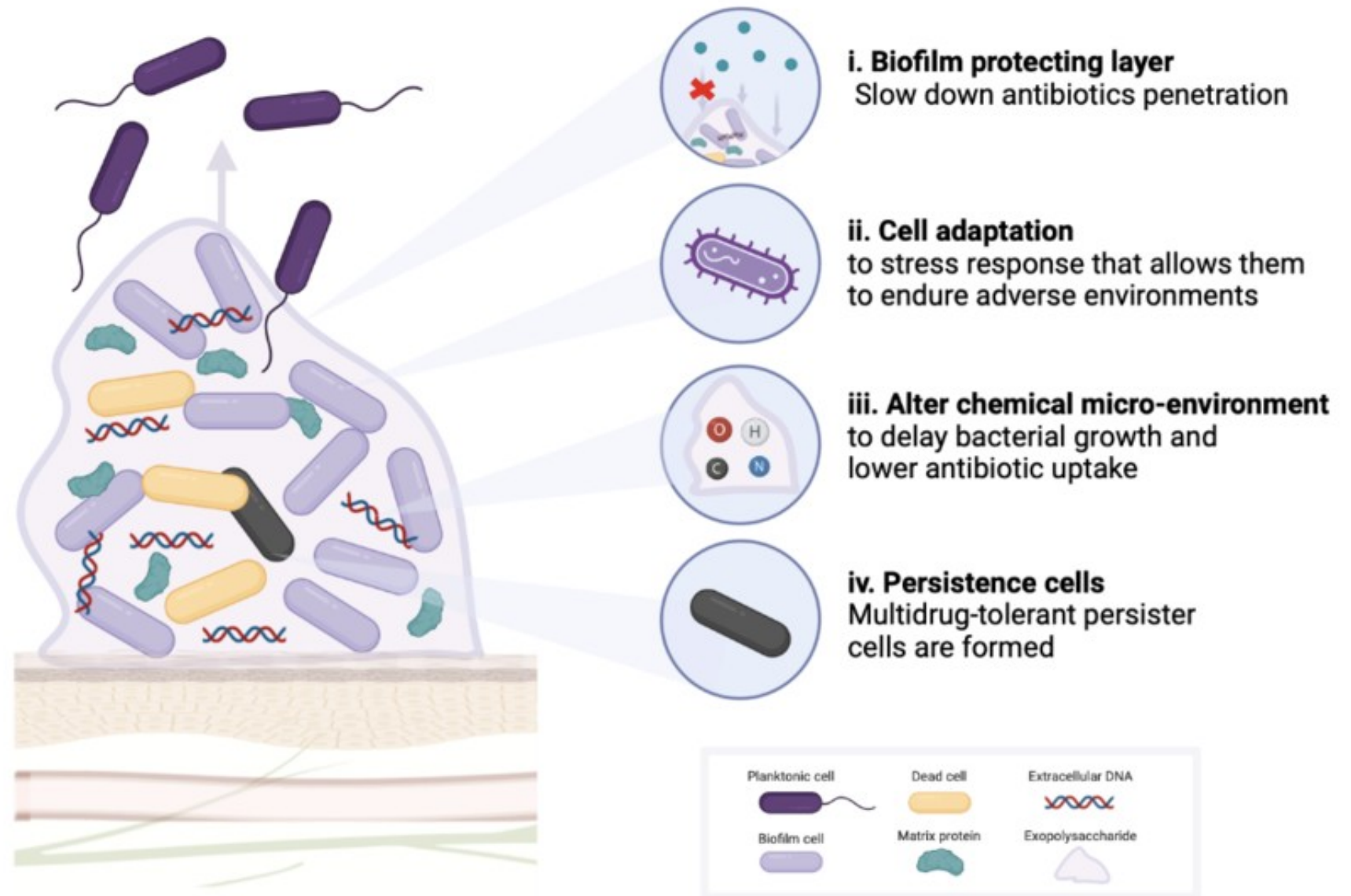
Mutation of WalK/R gene

walR → *walk*

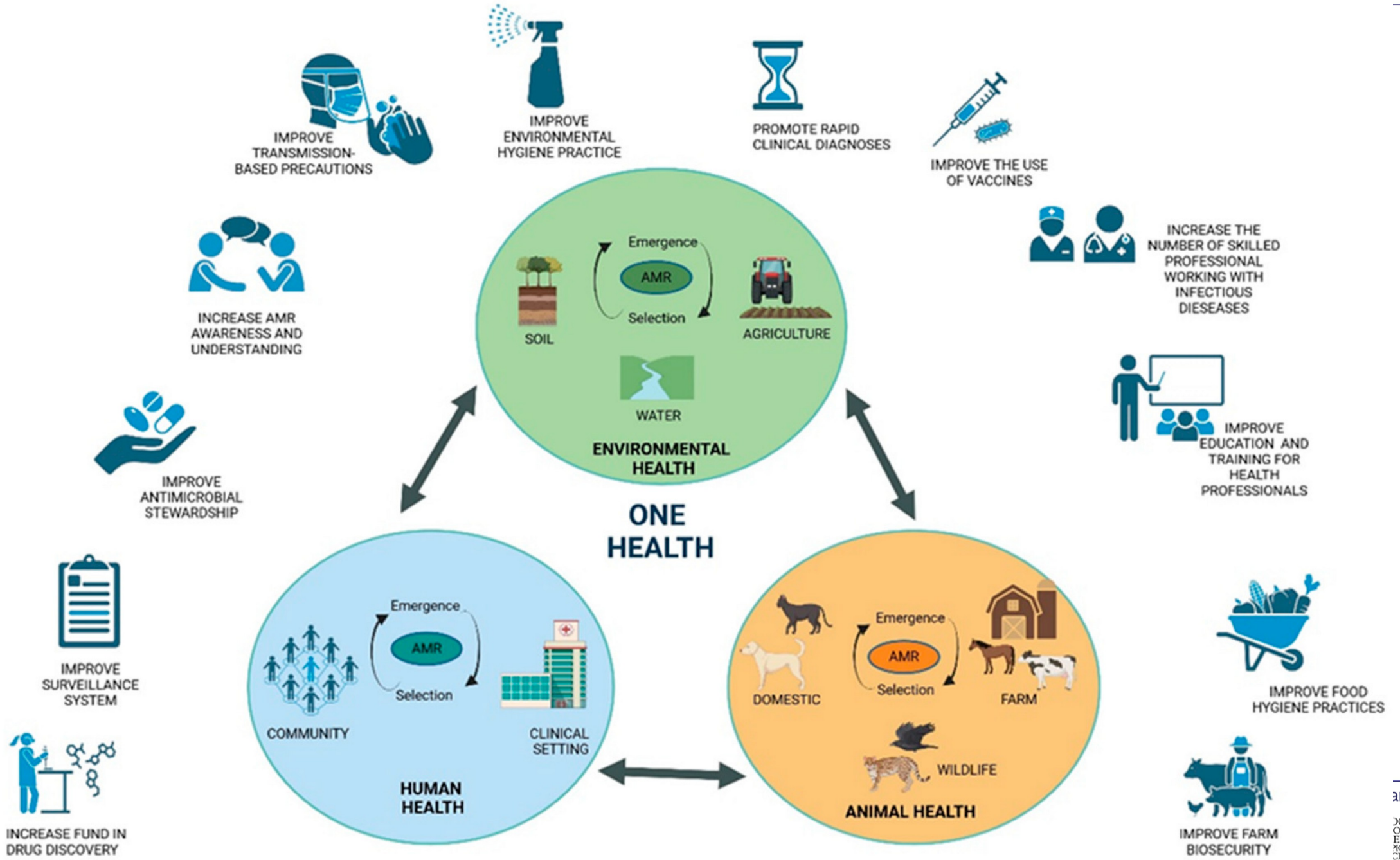


Mechanism 5: Biofilm-Mediated Resistance

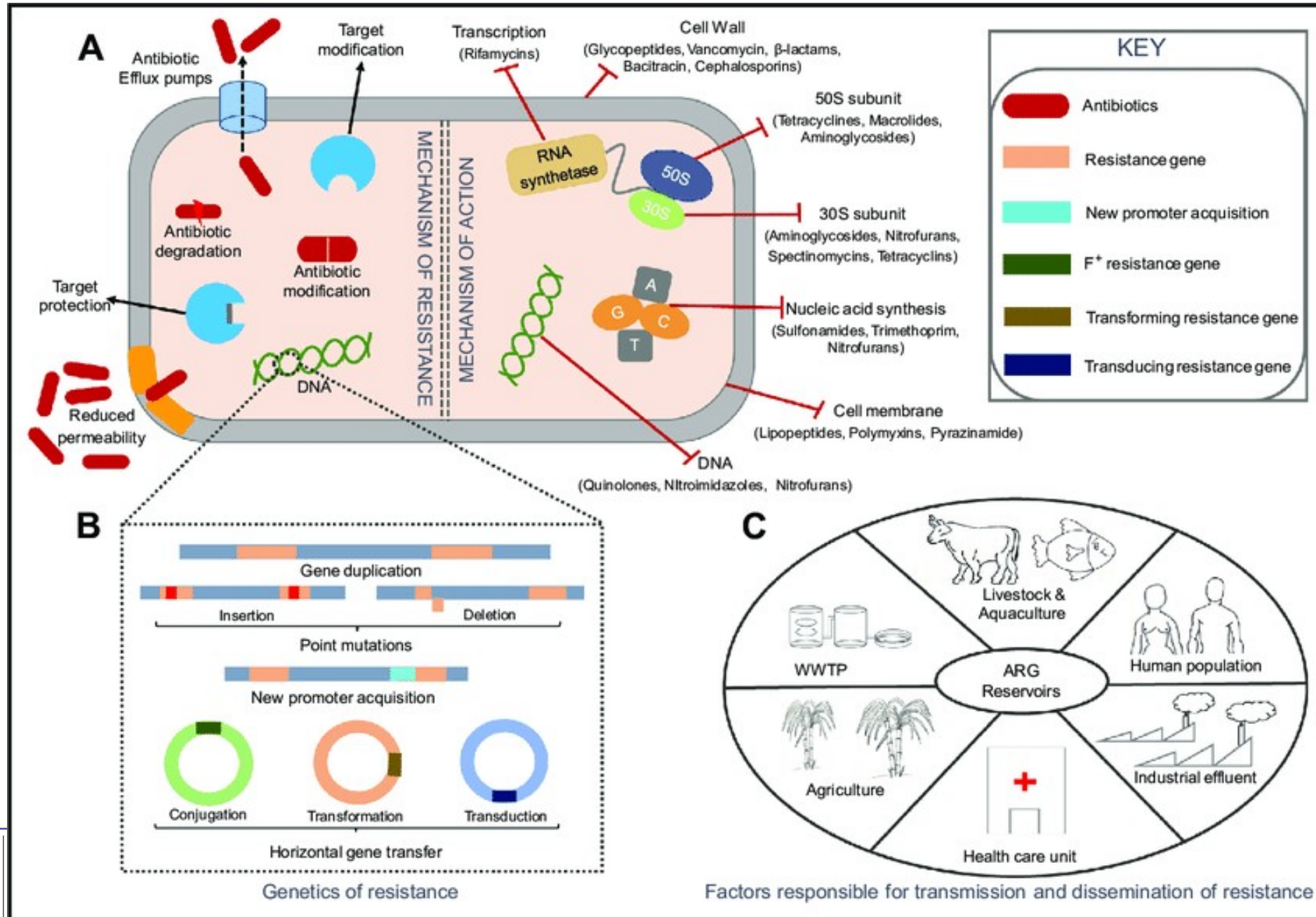
- Biofilms limit antibiotic penetration.
- Cells within biofilm adopt tolerant phenotypes (persisters).
- Clinical relevance: chronic infections, medical devices.



Application of AMR studies: One Health



Summary



Overview of Pathogenic Bacteria

- These are bacteria capable of causing disease in humans, often leading to significant health issues
- Through expression of virulence factors
- They can spread through contaminated food, water, direct contact with infected individuals
- Infections from these bacteria contribute to morbidity and mortality worldwide
- Understanding their genomic makeup helps in developing strategies to combat infections

Examples:

- ***Escherichia coli***; A diverse group with some strains being harmless while others can cause severe food borne illnesses
- ***Klebsiella pneumoniae***; known for its role in **hospital-acquired** infections and antibiotic resistance

Virulence factors in Pathogenic Bacteria

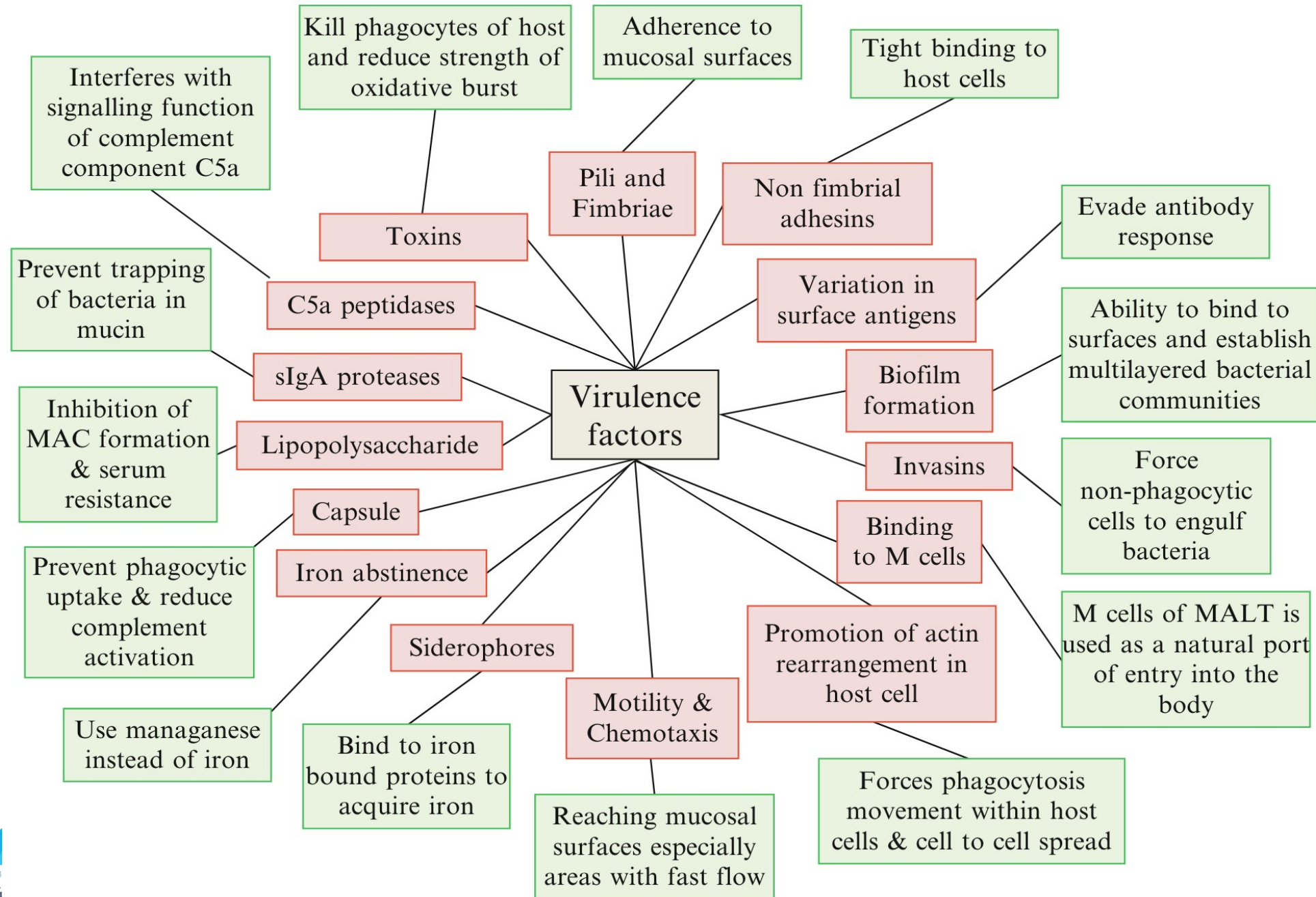
- These are molecules produced by pathogens that enable them to achieve colonization, evade host defenses, and cause disease
- Virulence gene sequences have abnormal amounts of **guanine** and **cytosine**

Examples

- **Toxins (exotoxins & enterotoxins)** disrupt host cellular processes and contribute to symptoms of infection
- **Adhesins** facilitate attachment to host cells, critical for colonization and infection establishment



Various virulence factors utilized by pathogens during infection



Mechanisms of action

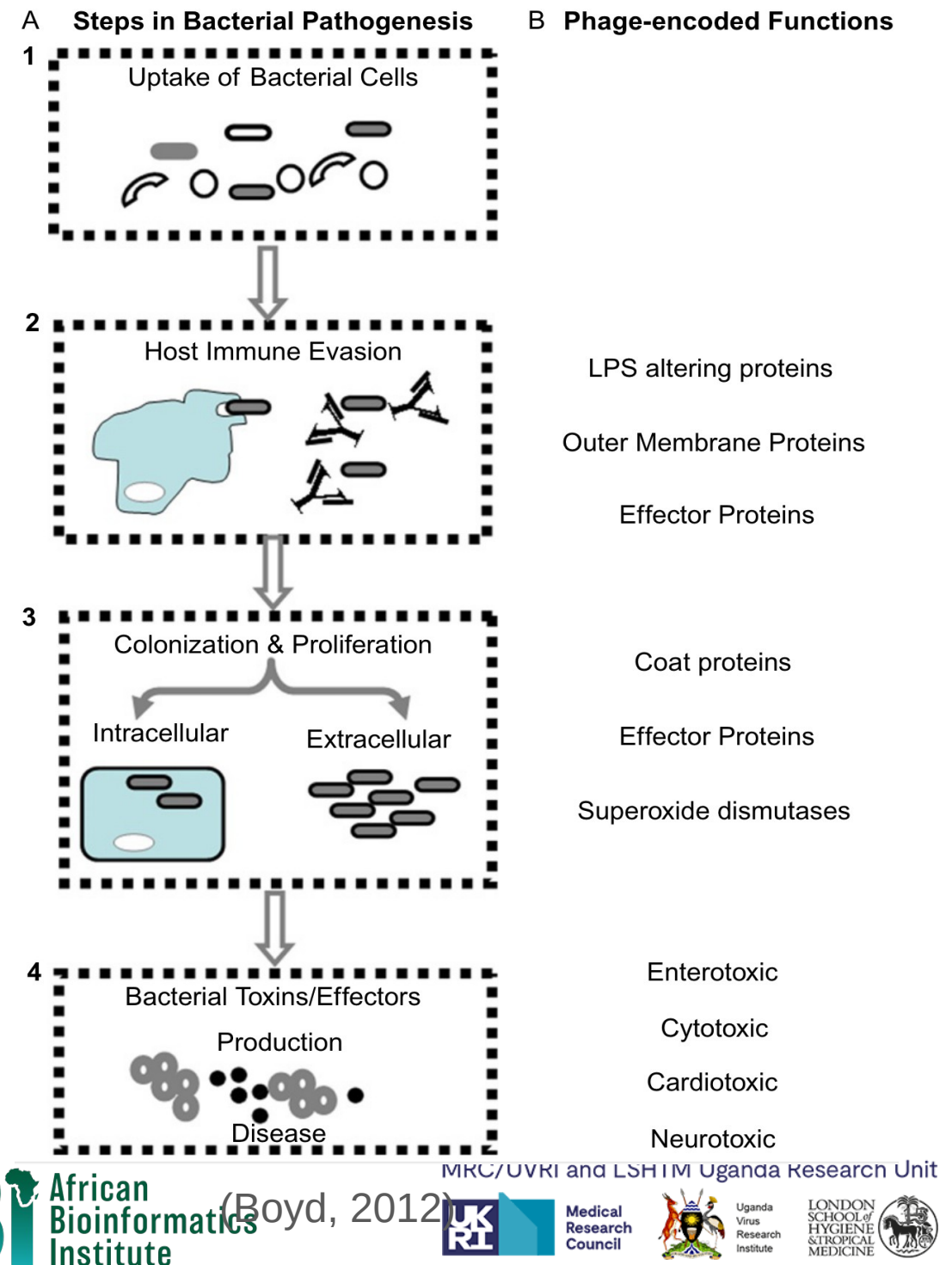
- Some virulence factors interfere with immune responses (e.g., evasion of phagocytosis)
- Others may alter host cell signalling pathways to promote bacterial survival and replication

Genetic Regulation

- The expression of virulence factors is often regulated by environmental signals, allowing bacteria to adapt to changing conditions within the host

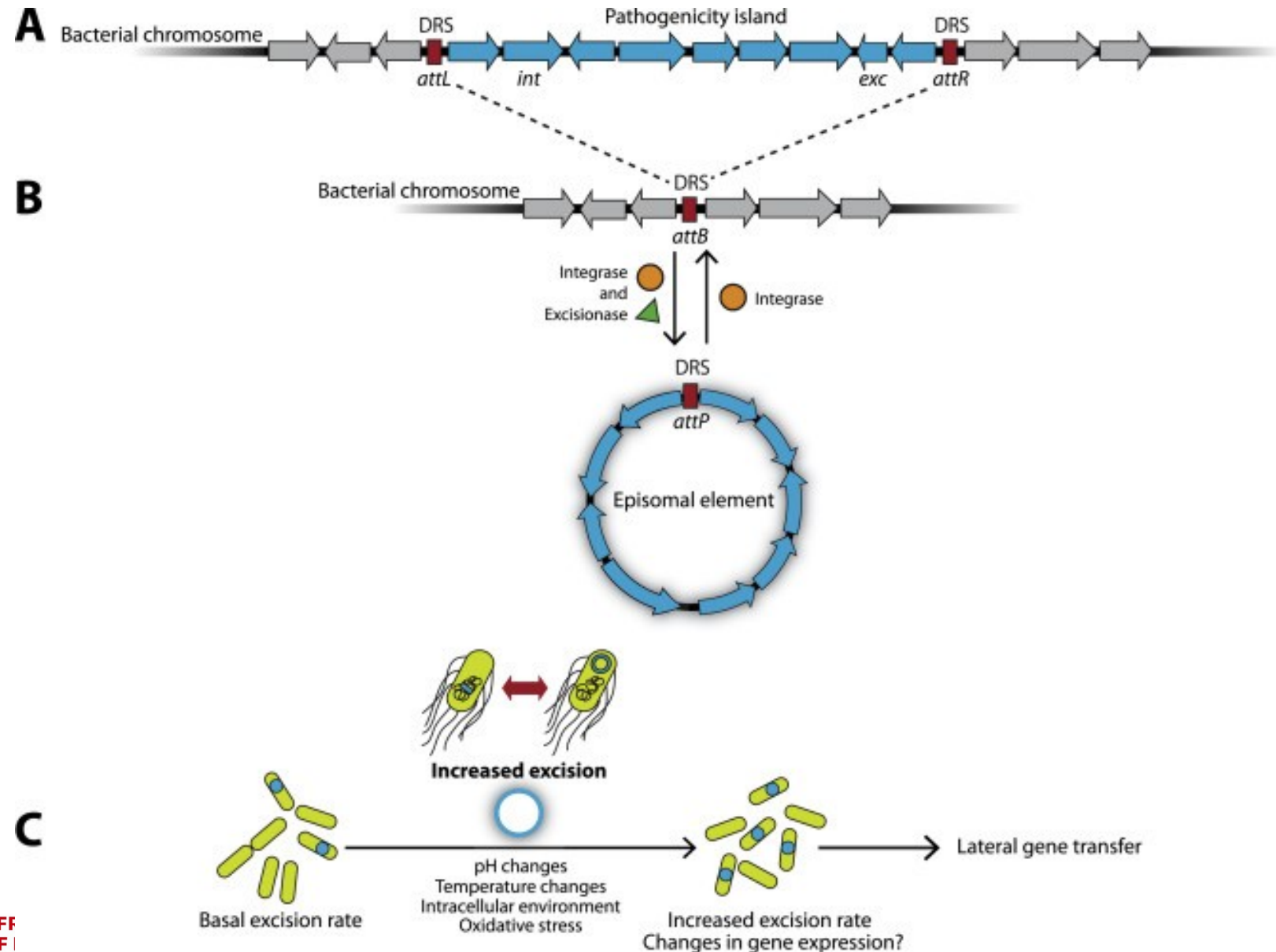
Research implications

- Understanding these factors help identify potential targets for vaccine development and new therapeutic strategies



Pathogenicity Islands

- These are large segments of DNA found within bacterial genomes that contain **clusters of genes** associated with virulence
- Found on a chromosome or separate plasmids
- The genes are for either **entry** or promoting **survival** on entry into the host



Pathogenicity Islands

Characteristics of PAIs

- Often flanked by tRNA genes or insertion sequences, indicating horizontal gene transfer origins
- Typically absent in non-pathogenic strains of the same species, highlighting their role in pathogenicity

Examples in *E.coli* & *K.pneumoniae*

- Specific PAIs encode genes for adhesins, toxins and virulence determinants essential for infection establishment and persistence

Clinical relevance

- The presence of PAIs can be used as a marker for virulent strains during epidemiological studies or outbreak investigations
- Understanding PAIs aids in developing diagnostic tools for rapid identification of high-risk pathogens

Case study 1 - *E. coli*

Genomic diversity; *E. coli* exhibits significant genomic variability; pathogenic strains can differ by up to 25% in their DNA sequences

Six categories cause human diarrhea

- Enterotoxigenic *E. coli* **ETEC**
- Enteroinvasive *E. coli* **EIEC**
- Enteropathogenic *E. coli* **EPEC**
- Enterohemorrhagic *E. coli* **EHEC**
- Enteroaggregative *E. coli* **EAEC**
- Diffusely adherent *E. coli* **DAEC**
- Uropathogenic *E. coli* (**UPEC**) - UTIs
- Neonatal meningitis-associated *E. coli* (**NMEC**)
- Toxins such as **Shiga toxin** disrupt cellular functions and lead to severe illness

Case Study 2 - *Klebsiella pneumoniae*

- *Klebsiella* is a Gram-negative rod, an opportunistic pathogen primarily affecting immunocompromised individuals,
- Associated with wide spectrum of hospital & community acquired infections (pneumonia & UTIs, intraabdominal, bloodstream infection, meningitis, pyogenic liver abscess)

Genomic insights:

- Virulence determinants; capsule, lipopolysaccharides, siderophores, type 1 & 3 fimbriae

Virulence factors

- **Capsule production** enhances evasion from the host immune response, making it more virulent
- **Biofilm formation** contributes to persistence on medical devices, complicating treatment efforts
- **Public health threat:** the rise of multidrug-resistant strains poses a significant challenge for treatment options globally
- **Future directions:** Research is focused on developing phage therapy as potential alternative treatment

Virulence Tools and Databases



- VFDB: curated virulence gene sets (core vs full datasets)



TRANSITIONS TO



- PATRIC / BV-BRC

- Kleborate



- Linux tools: Abricate + VFDB

Virulence Factor Database (VFDB)



-  **interface**
- [Acinetobacter](#)
 - [Aeromonas](#)
 - [Anaplasma](#)
 - [Bacillus](#)
 - [Bartonella](#)
 - [Bordetella](#)
 - [Brucella](#)
 - [Burkholderia](#)
 - [Campylobacter](#)
 - [Chlamydia](#)
 - [Clostridium](#)
 - [Corynebacterium](#)
 - [Coxiella](#)
 - [Enterococcus](#)
 - [Escherichia](#)
 - [Francisella](#)
 - [Haemophilus](#)
 - [Helicobacter](#)
 - [Klebsiella](#)
 - [Legionella](#)
 - [Listeria](#)

About VFDB

The virulence factor database (VFDB) is an integrated and comprehensive online resource for curating information about virulence factors of bacterial pathogens. Since its inception in 2004, VFDB has been dedicated to providing up-to-date knowledge of VFs from various medically significant bacterial pathogens.

The motivation for constructing VFDB was twofold:

- First, to provide in-depth coverage major virulence factors of the best-characterized bacterial pathogens, with the structure features, functions and mechanisms used by these pathogens to allow them to conquer new niches and to circumvent host defense mechanisms, and cause disease.
- Second, to provide current knowledge of the wide variety of mechanisms used by bacterial pathogens for researchers to elucidate pathogenic mechanisms in bacterial diseases that are not yet well characterized and to develop new rational approaches to the treatment and prevention of infectious diseases.

Definitions

- A **bacterial pathogen** is usually defined as any bacterium that has the capacity to cause disease. Its ability to cause disease is called **pathogenicity**.
- **Virulence** provides a quantitative measure of the pathogenicity or the likelihood of causing disease.
- **Virulence factors** refer to the properties (i.e., gene products) that enable a microorganism to establish itself on or within a host of a particular species and enhance its potential to cause disease. Virulence factors include bacterial toxins, cell surface proteins that mediate bacterial

Summary points

- Genomics provides critical insights into mechanisms underlying bacterial pathogenicity & resistance profiles among pathogens like *E. coli* & *K. Pneumoniae*
- Understanding virulence factors & pathogenicity islands is essential for developing effective vaccines & treatments against bacterial infections
- VFDB is the main curated reference resource
- Practical tools (Abricate, BLAST) allow detection
- Interpretation requires biological context